



Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-ID-PRT-6-Vers.1.0

Effective Date: 25/02/2024

Target Review Date: 20/02/2027

Title: Pediatric Tuberculosis: A Guide to the Gastric Aspirate Procedure

Introduction:

Gastric aspirate is an important tool for TB diagnosis in children as the children usually swallow their sputum.

This guide provides practical steps on how to collect and perform the gastric aspirate

Step 1:

Explain the procedure to the caregiver that it is a brief, temporarily uncomfortable to the child

A) The patient should be NPO after midnight.

B) The procedure should be as soon as the child awakens.

(Any family members who could possibly have active TB should wear masks to prevent transmission to healthcare workers and other patients.)

While young children with active TB are rarely contagious, treatments that elicit cough, such as stomach aspiration, should be performed in a well constructed facility with sufficient filtration and negative pressure. During the process, healthcare providers should use N-95 masks. The room should be closed for one hour, or as prescribed by your infection control provider / engineers.

Step 2:

- Measure distance the tube should be inserted into the stomach: by measuring the expected distance from nose to stomach. Stretch the tube from the tip of the nose, around the ear and down to the bottom of the Xiphoid process. This is the distance the tube should be inserted into the stomach

Step 3:

- Place NG tube into child's nose - stay away from the septum and aim directly perpendicular to the bed as you advance the tube

- look to make sure that the tube is not coiled in the mouth, at this age, children commonly vomit, so be prepared to collect any emesis.

- VERY RARELY, the tube will pass through the airway instead of the stomach. If the youngster has any respiratory discomfort or a muted cry. Remove the tube as soon as possible.

-A powerful unaltered cry, gastric contents in the tube, and a forceful gurgling sound auscultated over the stomach when air is put into the tube by syringe indicate successful tube placement.

Step 4:

- Early morning when the child a wake, before he/she eat, see or smell food: **aspirate the stomach contents with the syringe.**
- Place the gastric aspirates in a special bicarbonate-containing gastric aspirate tube or regular specimen cup
- If a special bicarbonate tube or cup is not available, the lab must neutralize the stomach acid with bicarbonate within 1/2 hour
- If less than 5 - 10 cc of mucous returns, re-position the tube and/or the child in order to look for the pool of mucous.
- If still < 5 - 10 cc of gastric contents have been aspirated; Prepare to instill water into the tube: **instill 20 - 30 cc of sterile water into the tube then quickly re-aspirate the content.**

Step 5:

- **Put all yield in sterile cup or tube.**
- **Immediately transport to lab to neutralize.**

OR

- **Neutralize at bedside:** *(Bicarbonate for neutralization - 2.5 grams NaHCO₃ dissolved in 100 cc deionized water. Filter the solution through a 45mm filter. Use 1.5 cc for each specimen. Lab should monitor and correct the pH)*
- **Order AFB smear and culture.**
- **Transport the specimen carefully and promptly to the lab.**

Reference:

- <https://npin.cdc.gov/publication/pediatric-tuberculosis-guide-gastric-aspirate-ga-procedure>

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